Team Details

Team name: Trinetra Labs

Team leader name: Rayhankhan Pathan

Team size: 1

Problem Statement: PS1 — Intelligent Conversational AI for the KSP Crime Database

Brief about the solution

TRINETRA — Tactical Real-time Intelligence & Network Engine for Tracking, Reasoning & Analytics

An agentic crime-intelligence console for the Karnataka State Police. An investigator asks a question in Kannada or English (voice or text); the AI agent queries the crime database, uncovers hidden criminal networks, profiles repeat offenders, and answers with an interactive graph, hotspot map and timeline — every fact cited to its FIR.

Days → seconds
FIR-to-lead cross-referencing

EN + ಕನ್ನಡ
voice & text, context-aware

100% on Catalyst
deployed, explainable, secure

Opportunities

a. How different is it from any of the other existing ideas?
b. How will it be able to solve the problem?
c. USP of the proposed solution

How it's different

- Not a dashboard and not a plain chatbot — an agent that returns understanding.
- Generative UI: it decides whether to draw a network, a map or a timeline.
- Relationships modelled as edge tables → graph in code (no graph DB needed).
- Kannada + English voice, built for Indian policing.

How it solves the problem

- Natural language → safe ZCQL over the Data Store, no SQL skills needed.
- Surfaces hidden offender / phone / money links across jurisdictions.
- Transparent risk scoring prioritises investigation.
- Cuts manual cross-referencing from days to seconds.

USP

- Agentic + explainable: every answer cites its FIRs (evidence trail).
- Spatiotemporal hotspots + organised-network detection in one console.
- Fully on Zoho Catalyst — Qwen LLM, Zia voice, Data Store, Slate.
- Production-grade: role-based access, audit-ready, scalable.

List of features offered by the solution

Conversational agent
Voice + text, English & Kannada, context-aware follow-ups.

NL → ZCQL retrieval
Plain questions become safe, validated Data Store queries.

Criminal network analysis
Hidden links between offenders, victims, phones, accounts.

Offender risk profiling
Explainable 0–100 score from weighted factors.

Spatiotemporal hotspots
District clusters layered with time-of-day.

Investigator timeline
Auto case chronology showing MO escalation.

Money-trail analysis
Suspicious transactions and mule-account links.

Similar-case leads
Find look-alike cases by MO — investigative leads.

Crime forecasting
Projected next-week hotspots for proactive patrol.

Early-warning alerts
Spikes, forming rings, repeat offenders.

Explainable & cited
Shows the generated ZCQL + reasoning; cites FIRs; PDF export.

Secure & governed
Role-based access, audit logs, SELECT-only query gate.

Process flow diagram or Use-case diagram

Investigator
asks in EN / ಕನ್ನಡ
(voice or text)

→

Agent
intent → generate ZCQL
→ safety gate

→

Data Store
validated ZCQL
over crime DB

→

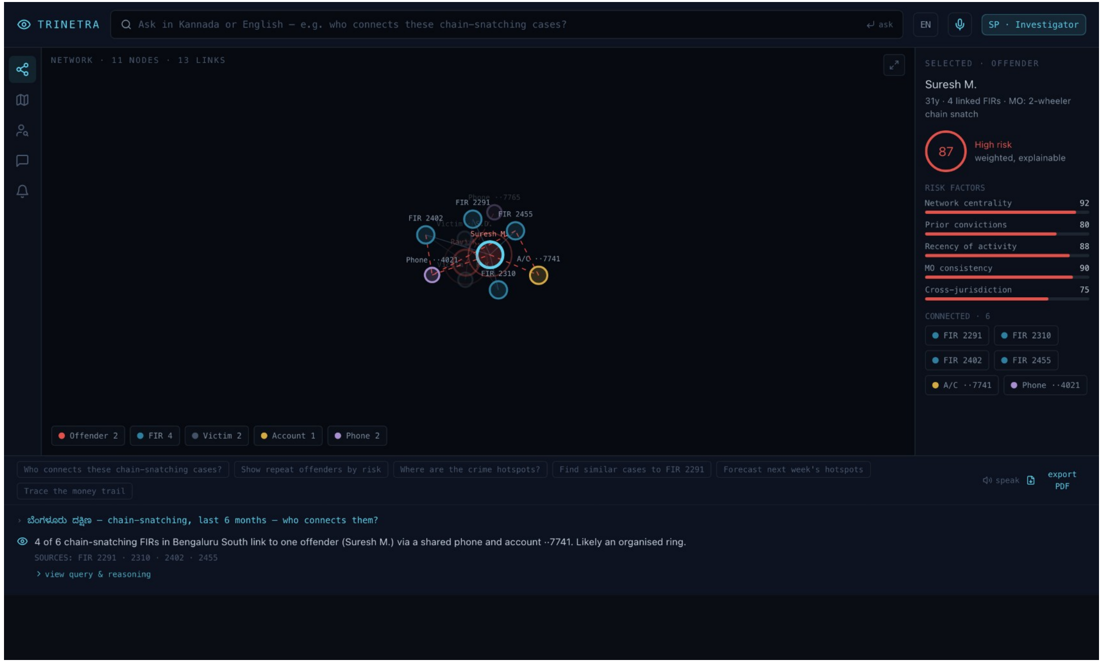
Analysis
network · risk ·
hotspot · money

→

Answer
graph / map / timeline
+ cited FIRs · PDF

Use case: “Who connects these chain-snatching cases?” → the agent runs the query, finds 4 FIRs tied to one offender via a shared phone + mule account, and renders the ring with an evidence trail — in seconds.

Wireframes/Mock diagrams of the proposed solution (optional)



Console layout

- **Command bar —** natural-language + voice query
- **Network canvas —** force-directed graph, neighbour highlight
- **Inspector —** risk factors + connected entities
- **Agent log —** cited answer · view query & reasoning · PDF

Architecture diagram of the proposed solution

Frontend — Catalyst Slate (Next.js console: chat · network graph · hotspot map · timeline)

↓

API Gateway → Functions: agent (NL→ZCQL) · network-build

↓

QuickML
Qwen LLM + RAG

Zia
voice EN/KN

Data Store
ZCQL + edges

Stratus
files

SmartBrowz
PDF

Auth
RBAC + audit

Synthetic data → Data Store. Every generated ZCQL is validated (SELECT-only, whitelisted tables) before execution — the governance gate.

Technologies to be used in the solution

Frontend
Next.js 14 · React · TypeScript · Tailwind CSS · react-force-graph-2d · MapLibre GL · Web Speech API

Backend / Functions
Node.js Catalyst Functions (Advanced I/O) · ZCQL · REST + API Gateway

AI / ML
Qwen via Catalyst QuickML LLM-Serving · RAG knowledge base · Zia STT / TTS / translation

Data
Catalyst Data Store (relational + edge tables) · Python synthetic-data generator (planted rings)

Platform
Zoho Catalyst — Slate hosting · Authentication · Stratus · SmartBrowz · Cron · Signals

List down Catalyst Services being used in the solution

Slate / Web Hosting
Hosts the Next.js console

Functions (Advanced I/O)
Agent (NL→ZCQL) & network-build logic

API Gateway
Routing, throttling, auth in front of Functions

QuickML — LLM Serving + RAG
Qwen for NL→ZCQL & grounded answers

Zia Services
Kannada/English speech-to-text, TTS, translation

Data Store (ZCQL)
Crime DB + relationship edge tables + search

Stratus
Case files, evidence, exported PDFs

SmartBrowz
Conversation → PDF report generation

Authentication
Role-based access (investigator/analyst/supervisor)

Cron + Signals
Scheduled risk recompute & early-warning alerts

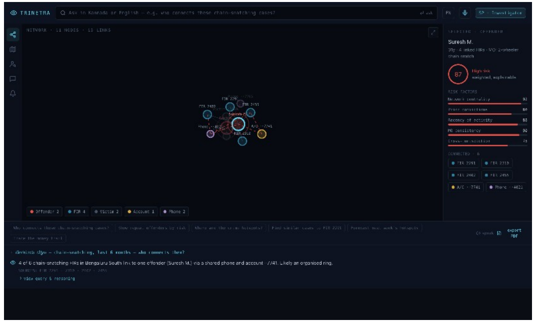
Estimated implementation cost (optional)

Built within the hackathon Catalyst credits (₹1,500 + $250). Indicative monthly run-rate at SCRB pilot scale:

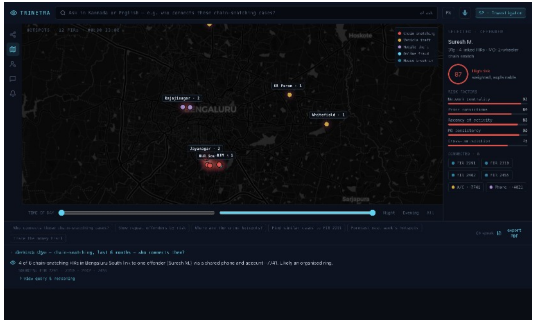
Catalyst service	Usage	Indicative / month
Functions (agent + network)	~50k invocations	Within free tier → low
Data Store	~2–5 lakh records	Low
QuickML LLM-Serving (Qwen)	~30k queries	Usage-based
Zia voice	~5k minutes	Usage-based
Slate + Stratus + SmartBrowz	hosting + PDFs	Low

Snapshots of the prototype

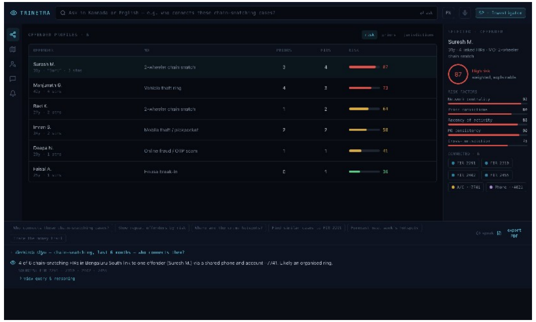
Network — organised ring



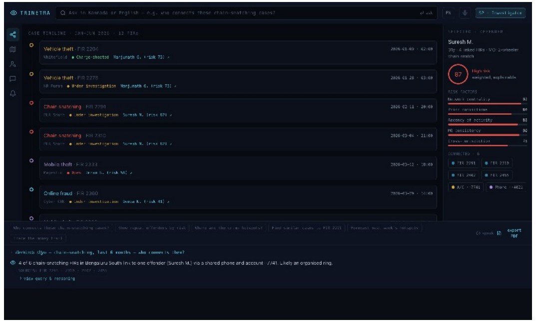
City hotspot map (MapLibre)



Offender risk profiles



Case timeline



Prototype Performance report/Benchmarking

< 1s
agent response (mock mode)

100%
malicious queries blocked by ZCQL gate

1.5 lakh
FIRs generated & queried in tests

6/6
demo intents return grounded, cited answers

16 nodes
network built per offender ego-graph

11 tables
relational schema, edge-modelled

Provide links to your:

1. GitHub Public Repository
2. Demo Video Link (3 Minutes)
3. Deployed Link

GitHub repository: https://github.com/mrayhankhan/trinetra

Demo video (3 min): https://youtu.be/<unlisted-link>

Deployed link (Catalyst): https://trinetra-60074156791.development.catalystserverless.in/app/index.html

Replace with your public links before submitting. Deployment must be on Catalyst.

Additional Details/Future Development (if any)

- **RAG criminology knowledge base** — Ground answers in IPC / case-law / SOP documents via QuickML RAG.
- **Crime forecasting** — QuickML time-series + Zia AutoML to predict emerging hotspots.
- **Financial-crime expansion** — Deeper money-mule and cross-bank transaction graphs.
- **More Indic languages** — Beyond Kannada — Hindi, Tamil, Telugu, Urdu.
- **Mobile + field app** — Push alerts and on-the-go queries for officers.
- **Live SCRB integration** — Connect to real KSP systems with full governance & DPDP compliance.

Karnataka State Police
Government of Karnataka

Powered by H2S

Datathon 2026

Technology partner ZOHO

THANK YOU